

```
''' Data Visualization III '''
```

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

```
import seaborn as sns
import matplotlib
%matplotlib inline
```

```
df = pd.read_csv("IRISDATA.csv", header=None)
df.columns = ['col1', 'col2', 'col3', 'col4', 'col5']
```

```
df.head()
```

	col1	col2	col3	col4	col5
0	5.1	3.5	1.4	0.2	Iris-setosa
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa
3	4.6	3.1	1.5	0.2	Iris-setosa
4	5.0	3.6	1.4	0.2	Iris-setosa

```
column = len(list(df))
column
```

```
5
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
#   Column  Non-Null Count  Dtype
---  -
0   col1    150 non-null         float64
1   col2    150 non-null         float64
2   col3    150 non-null         float64
3   col4    150 non-null         float64
4   col5    150 non-null         object
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
```

```
np.unique(df['col5'])
```

```
array(['Iris-setosa', 'Iris-versicolor', 'Iris-virginica'],
      dtype=object)
```

```
fig, axes = plt.subplots(2,2, figsize = (16, 8))
```

```
axes[0,0].set_title("Distribution of first column")
axes[0,0].hist(df["col1"]);
```

```

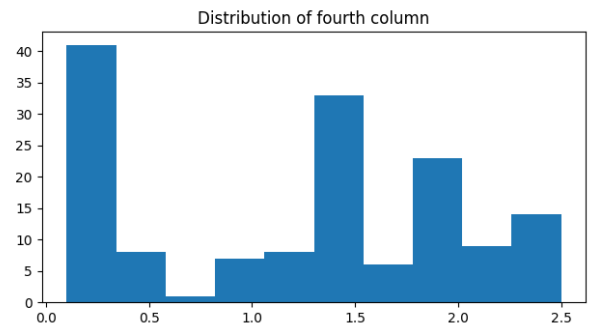
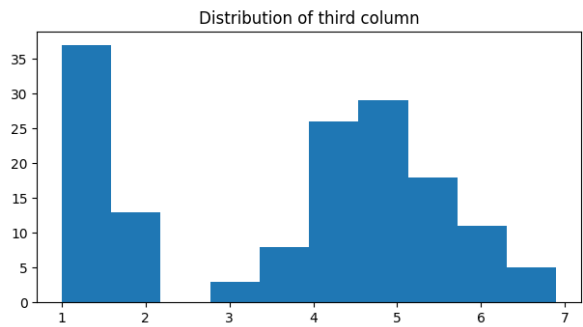
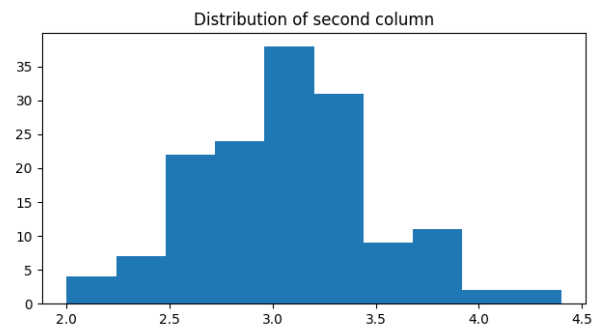
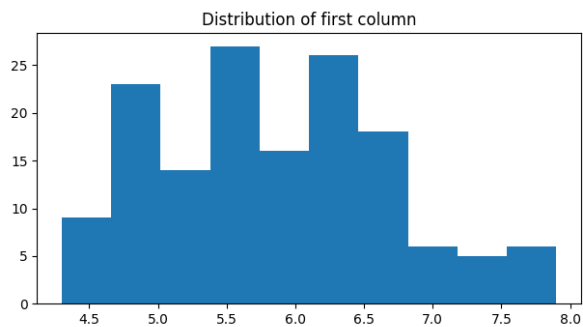
axes[0,1].set_title("Distribution of second column")
axes[0,1].hist(df["col2"])

axes[1,0].set_title("Distribution of third column")
axes[1,0].hist(df["col3"])

axes[1,1].set_title("Distribution of fourth column")
axes[1,1].hist(df["col4"])

(array([41., 8., 1., 7., 8., 33., 6., 23., 9., 14.]),
 array([0.1 , 0.34, 0.58, 0.82, 1.06, 1.3 , 1.54, 1.78, 2.02, 2.26,
2.5 ]),
<BarContainer object of 10 artists>)

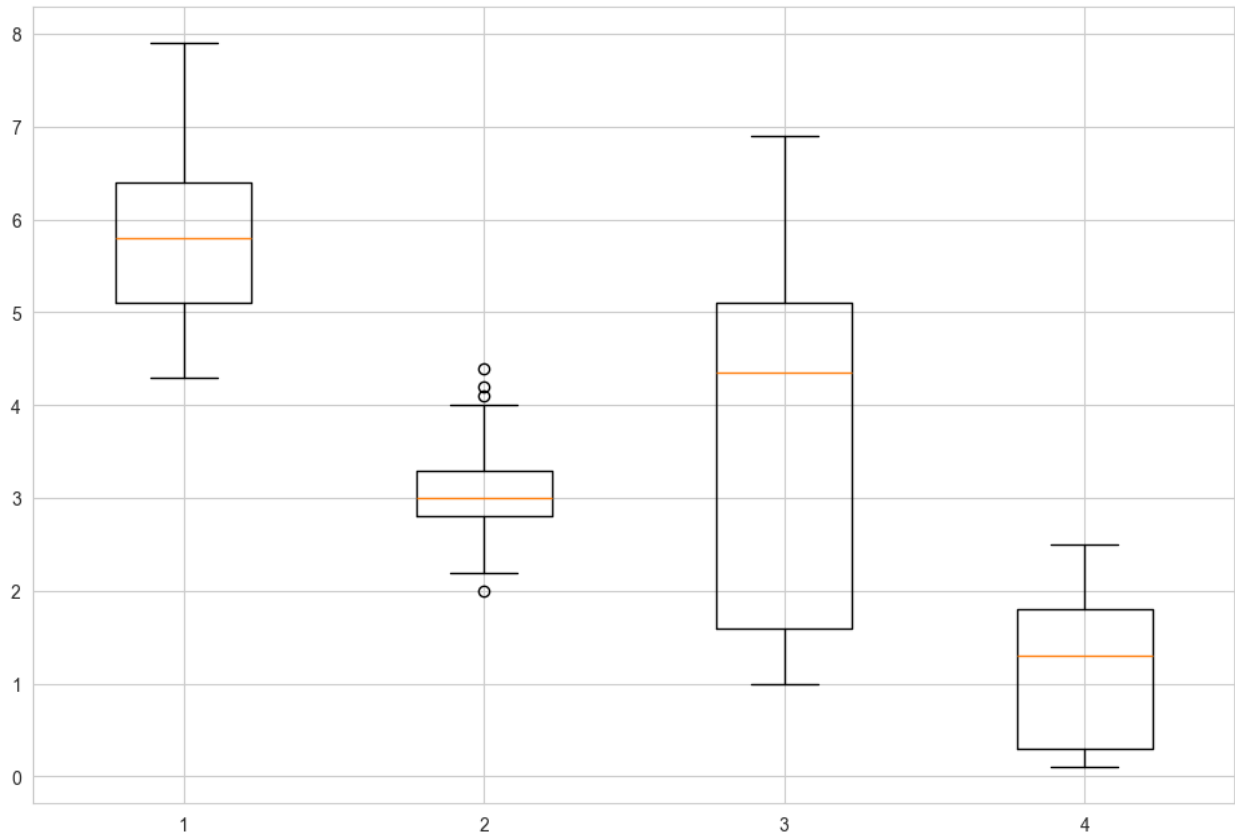
```



```

data_to_plot = [df["col1"],df["col2"],df["col3"],df["col4"]]
sns.set_style("whitegrid")
fig = plt.figure(1,figsize=(12,8))
ax = fig.add_subplot(111)
bp = ax.boxplot(data_to_plot);

```



```
df.describe()
```

	col1	col2	col3	col4
count	150.000000	150.000000	150.000000	150.000000
mean	5.843333	3.054000	3.758667	1.198667
std	0.828066	0.433594	1.764420	0.763161
min	4.300000	2.000000	1.000000	0.100000
25%	5.100000	2.800000	1.600000	0.300000
50%	5.800000	3.000000	4.350000	1.300000
75%	6.400000	3.300000	5.100000	1.800000
max	7.900000	4.400000	6.900000	2.500000